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A life cycle assessment of the environmental impact of children's toys

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ABSTRACT

Quantifying product impact is one way to make the environmental consequences of purchases clear to consumers and retailers. This paper provides a Life Cycle Assessment (LCA) of the environmental impacts of common children's toys. We compare 3 Lego™ sets, 1 Barbie™, 1 Jenga game, 1 plush dog, 1 plush dog with battery components, and 1 Marble Frenzy™ game, each representing a larger category of toy. After determining the materials in each toy, we built these toys in OpenLCA with existing materials data from LCA databases. Our results showed that a Lego Star Wars set had the highest eutrophication potential with the largest single contribution to impact from injection molding. This Lego set also had the highest GHG (Greenhouse Gas) emissions due to the use of acrylonitrile butadiene styrene (ABS) plastic. Jenga had the highest acidification impact but the lowest GWP of all the toys despite having the highest mass. Our results indicate low GHG emissions and eutrophication potential of wood as a toy material. Wood's acidification potential, however, was the highest of all studied toy materials since impacts included land management for forestry though final wood processing. While plastic had a higher impact per mass than wood, the type of plastic used was important in determining the GHG emissions: ABS and polyvinylchloride (PVC) composition was lower impact than nylon granulate, and other plastics. Since our functional units determined impact per twenty hours of use over 2 years, increased toy longevity was one option to reduce toy impact, though even increased toy longevity should be combined with design for sustainability coupled with transparent environmental labeling to communicate the environmental value of low impact toys to consumers.

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1. Introduction

In 2019, US consumers spent $20.91 billion dollars on toys, including: action figures, dolls, puzzles, plush toys, and vehicles, among others (Statista, 2020). While the research associated with toys is extensive, it is mainly limited to safety (Aurisano et al., 2021), consumer preferences, (Larson and Jordan, 2019; Perez-Belis et al., 2015; Richards et al., 2020; Scherer et al., 2017) and play behavior (Isenberg and Quisenberry, 2002; Ishibashi and Uehara, 2020; Wang et al., 2020), though see (Charmondusit et al., 2014; Margallo et al., 2021; Rangaswamy et al., 2018). While the materials that comprise toys have been researched as part of other work to quantify the impact of material production and recyclability (Chen et al., 2021; Faraca and Astrup, 2019), there is limited research on the specific environmental impact of individual toys (Margallo et al., 2021; Muñoz et al., 2009) and even less on comparisons between toy product types (Rangaswamy et al., 2018; Watkins and Mestre, 2019).

Research on toy impact is important both in the design phase to reduce the impact per toy (Margallo et al., 2021), but also for consumers. Most toys are designed for short term use, then disposal (Johnson, 2001). In France, 40 million toys end up in landfills annually (Robertson-Fall, 2020) due, in part, to their design for this fate. Toys can be made of materials with more longevity (Watkins and Mestre, 2019), repaired, or designed for end-of life recycling. Even when made of recycled materials like plastics, however, their complicated forms and chemical additives cause toys to be rated as medium to low recyclability in facilities (Faraca and Astrup, 2019) leading to landfill disposal. Indeed, nearly 80% of all toys end their life cycles in landfills, incinerators, or the ocean (Robertson-Fall, 2020) and toys make up almost 6% of landfill plastics (Ejlertsson et al., 2003). The growing popularity of toys with electronic components means that toys can also contribute to e-waste streams (Watkins and Mestre, 2019).

Some research has investigated consumer preferences for eco-alternatives like bioplastic sand toys (Scherer et al., 2017), and other research has focused on the environmental impact of changes in toy design. Choi et al. (1997) conducted an improvement analysis of a toy train. They found that changing the wheel manufacturing from machining to die-casting resulted in a 28% de-

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crease in energy required for production. Improved product design further decreased the environmental burden of toy train production. Other research has measured the eco-efficiency of wooden toy companies by using economic and environmental indicators, including net sales, gross margin, material, energy, water consumption, and waste disposal (Charmondusit et al., 2014). A comparative LCA indicated that locally-made *Channapatna* toys were less toxic, consumed less energy, and had minimum environmental impacts compared to PVC toys made in China (Rangaswamy et al., 2018). A detailed LCA investigated the environmental impacts of an electronic encyclopedia. Strategies to reduce impact were identified, proposed to the company, and developed in an eco-designed version of the encyclopedia that achieved approximately 50% reduction in all impact categories associated with production. Improvement analysis was also used to modify design of a teddy bear with electronic components (Muñoz et al., 2009).

Quantifying product impact also is one way to make the environmental consequences of purchases clear to consumers and retailers. Eco-labels that tout a product's environmental benefits are becoming more common for many food and household products (Bjorner et al., 2004; Carlsson et al., 2007; Mahe, 2010; Nicholls, 2002; Teisl et al., 2002). From the retailer perspective, labeling can be a way to differentiate based on quality (Sepúlveda et al., 2016) or responsible sourcing (Andorfer and Liebe, 2012; Rousseau, 2015). Environmental impact quantification can also help with initiatives like carbon labels that make more complete information available for consumers (Klimas and Shaffer, 2019). To date, there is no universally adopted method to advertise and compare the quantifiable environmental impacts of children's toys. This paper is a step toward this goal. We compared the environmental impact of eight toys across a range of toy categories (i.e. games, dolls, assembly sets) so that consumers can make informed decisions when purchasing toys. While sustainable toy purchasing can take on many forms including purchasing second-hand, hand-me-downs, purchasing durable toys, or recycling, this paper focuses specifically on quantifying the impact of selected toys within larger categories of toys and then discussing how the materials, transportation, and toy use contribute to the overall toy impact. The objective of this study is to help inform conscious consumers as they consider toy purchases. This study is not meant to recommend a *right* toy choice, rather to look at impact hot spots in toy construction that could help consumers think about their toy purchases.

2. Literature review: the importance of play in child development

Play was identified as instrumental in child development in Lev Vygotsky's seminal 1978 Developmental Psychology paper, "*Play and its role in the mental development of the child" (as found in Vygotsky, 2016).* Studies on play continue to the present day and include stakeholders from: The American Academy of Pediatrics, toy companies such as The Lego Foundation (Whitebread et al., 2017), and play-focused researchers (Brussoni et al., 2012; Moller, 2015; Yogman et al., 2018a; Yogman et al., 2018b). Toys play an integral role in child development (Brussoni et al., 2012) by transforming abstract ideas into concrete objects, thus helping children connect concepts (Moller, 2015). Toys can also teach children to be environmental stewards (Barreto et al., 2013) as seen in a case study from Hong Kong that used familiar toys to teach children about conscious consumption (Wong, 2018a). Teachers began with teaching 3- and 4-year-olds about refusing plastic and gave 5-year-olds the opportunity to design toys that could be shared between all children. Children designed two toys from reusable materials: (1) a football board game made from a shoebox, disposable chopsticks, and old clothes pegs, and a paper football; and (2) a vending machine created from boxes where turning a toilet roll released eggs filled with toy surprises (Wong, 2018b).

Toys are essential to the learning and development of children and different toys can impact early childhood development by providing outlets to practice fine and gross motor skills, and facilitate imaginative play by providing children a physical object to share (Wong, 2018b). While many toys have a positive impact on learning, toys can vary from high environmental impact technological toys like computers (Duan et al., 2009) to low impact plush animals (see result in this paper). A toy's environmental impact must be understood by parents and teachers to help promote sustainable systems that foster early childhood education alongside environmental stewardship (Dündar, 2017), and values (Dinella and Weisgram, 2018).

This study recognizes the importance of play for child development and education while focusing on the understudied issue of toy sustainability. To address this gap, this paper will compare a subset of toys to analyze their environmental impact.

By analyzing the impacts of various toys using life cycle assessment methodology, we can compare the impacts of each toy and break down impacts per material to potentially encourage conscious consumption and inform design. Traditional methods of conscious consumption revolve around third party certifications such as some ecolabels (Galarraga Gallastegui, 2002) Fair Trade (Jaffee and Howard, 2015) and a few token movements ("chemical free", "natural fibers", etc.). While the presence of labels is understood to be an improvement relative to no labels, consumers have difficulty distinguishing between the benefits associated with different labels (De Pelsmacker et al., 2005b; Rousseau, 2015), and it is unclear if labels cause buying behavior changes (Brach et al., 2018; Carrington et al., 2010). With more transparency associated with the environmental impacts of toys we hope to make it easier for environmentally conscious adults to make better-informed toy purchase decisions.

3. Methods

3.1. Life cycle assessment

Life cycle assessment (LCA) is a technique used to analyze the environmental impacts associated with all the stages of a product's life from material raw extraction through disposal or recycling (United Nations Environment Program, 1996). LCA is the most objective and frequently used quantitative tool used to calculate the environmental impact of products and services (Garcia-Herrero et al., 2017). Inputs in each stage like raw material selection, manufacturing energy, and transportation distance have associated impacts. By converting dissimilar inputs to similar environmental impact categories, LCA allows for comparison of impact between product stages (e.g. cotton production and transport) and between products (e.g. toys). Using LCA, various elementary flows (i.e. LCI inputs and outputs to and from nature) are converted into environmental impact categories using characterization factors. We used CML baseline to convert emissions to impact categories created by the University of Leiden in the Netherlands in 2001, CML baseline contains the most common impact categories used in LCA (Institute of Environmental Sciences CML, 2016). Ecoinvent was preferred as a source for materials data due to its reputation as the world's most consistent and transparent life cycle inventory database and the fact that it includes products and processes worldwide (Wernet et al., 2016; EcoInvent Center, 2022). When a material was not found in Ecoinvent, Gabi was used as the next preferred source. For transportation, since Ecoinvent did not have U.S. vehicle emissions and the majority of truck transportation was occurring in the U.S., we used vehicle data from the

National Renewable Energy Laboratory (NREL) for truck transportation.

3.2. Goal and scope

The main objective of this paper was to evaluate and compare the environmental impact of eight toys. The audience for this study is consumers interested in more environmentally responsible purchasing. Our methodology is transferable to any comparison study of consumer goods.

3.3. Functional unit

Functional units specify the service an object provides and the duration of that service. For this study, we defined the service as a toy (service) that provides at least 20 h of accumulated entertainment over a two-year period. This is similar to the functional unit described for toys in a recent LCA study (Margallo et al., 2021). The 20 h may be an underestimate (or an overestimate); the actual frequency of use depends on both the toy and the child. Additionally, many of the toys selected for this study are passed down between siblings, family members, and friends, extending the duration of their entertainment provision.

3.4. Toy descriptions

Our study included a small subset of children's toys that included: 3 Lego™ sets, 1 Barbie™ (based on the intermediate mass example from 3 dolls of varying body types in Table 1), 1 Jenga™ game, 1 plush dog, 1 plush dog with battery components, and 1 Marble Frenzy™ game. Each of these toys either represented large sales in their sectors or was a sample of a larger toy category. Each toy's mass is recorded in Table 1.

- Barbie is the largest single player in the plastic doll sector. Approximately 60 million Barbie dolls are sold annually (Daspin, 2020) and over 92% of American girls between the ages of 3 to 12 have owed a Barbie (Daspin, 2020).
- Lego is a large toy company: in 2008 Lego sold 7 Lego sets every second amounting to over 220 million sets in a single year (Diaz, 2008).
- A classic game, in 2009 Jenga sold over 2 million Jenga sets (as referenced in Gwyther, 2009).
- In America alone, over $1 billion worth of plush toys were purchased in a single year (O'Connell, 2020).

3.5. Transportation

Transportation emissions factored into each toy's overall environmental impact. We assumed that toys travelled from the manufacturing site in Guangzhou, China to the distribution site in Chicago, IL USA. We chose Guangzhou as the default manufacturing location as "China manufactures about 70 percent of the world's toys and is home to more than 8000 toy firms, most of which are located in the Pearl River Delta of Guangdong Province" (Chen et al., 2016). Several of our toys were manufactured in multiple locations. Barbie was manufactured in Nuevo Leon, Mexico and Guangzhou, China and Lego was manufactured in Billund, Denmark, Nyíregyháza, Hungary and Monterrey, Mexico. For the initial analysis, we centered Barbie production in China and ran a scenario comparison to determine the difference in product impact that resulted from switching production to Mexico. Since most Lego toys were manufactured in Denmark (LEGO 2021), we centered all Lego production in this area.

Due to a lack of public data regarding exact locations of Chinese toy manufacturing plants, we utilized Google Earth to find a section of plants nearby the Guangzhou shipping port and recorded 27.35 km from plant to port (Robertson and Klimas, 2019). Next we loaded our toys on a shipping container and transported them by boat 153.72 kms from Guangzhou port to Da Chan port in Shenzhen due to the high volume of shipping traffic that moves through the Da Chan port (Liu et al., 2013). We assumed that all toys, except for Lego and Jenga, would be transported 19,087.16 kms by boat from Shenzhen (Da Chan Port) to the US though the Los Angeles port. Toys were then transferred to a long-haul truck at the LA port for their 3268.58 km transport to Chicago, IL. We chose Chicago as the toy's final destination because Chicago is central in the United States and serves as a large hub for finished material distribution. Lego was transported from its Billund, Denmark factory 263.93 km to Copenhagen by truck then 6502.295 nautical miles (12,042.25 km) to New York harbor. The Lego sets then moved by truck 1276.85 km from New York, NY to Chicago, IL. Jenga was moved from Nuevo Leon, Mexico 2467.12 km to Chicago, IL by truck.

To determine transportation impacts, we simulated the packing of a shipping container based on each toy's mass and dimensions. We packed each shipping container with one type of toy using 36″ by 36″ by 36″ cardboard boxes. A 53′ Truck fits 102 of these cardboard boxes. Utilizing this method, we were able to determine how many of each type of toy would travel on a shipping container. For example, 112 plush dogs with a battery fit in one 36″ by 36″ by 36″ cardboard box. The 112 plush dogs with battery multiplied by the 102 cardboard boxes equals 11,424 plush dogs per shipping container. The only exception to this is Jenga which was packed in an 18″ by 18″ by 18″ box due to its larger mass. Once we determined how many nautical and land miles the shipping container traveled and the mass of the loaded shipping containers, we calculated the kg*km for the entire shipping container. Then we divided the total shipping container impact by the number of toys per shipping container to determine individual toy transportation impact. The volume of the shipping container for toy transport was kept the same between all toys regardless of shipping location or transportation method.

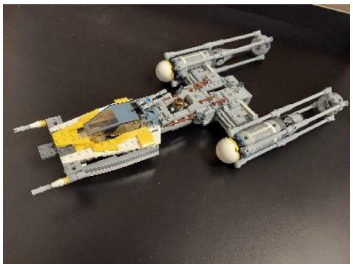
3.6. Material differentiation

When we could not find data on the material type from the manufacturer, we used existing resources to differentiate between plastics and other toy materials. We utilized Consultek's "Plastic and Elastomers Identification Chart" to best identify plastic types and assign corresponding product systems in OpenLCA (Ciroth, 2007). We conducted experiments to test if the unknown plastics sank or floated and determine what color flame arose while burned. We determined that the marbles of Marble Frenzy were polyvinyl chloride (PVC) by burning the sample with a flame over 500°F (plastic softened), dropped the sample in water (sank), burned a small corner of the sample (self-extinguished), tested if the sample dripped after burning (no). We also took trips to the local craft store to learn how to differentiate between fabrics: fleece, felt, polyester, polyester blends, etc.

3.6.1. Plush dogs (with and without batteries)

Both Plush Dogs (with and without batteries) were disassembled to identify the material components (Table 2): plastics (including fibers), and the battery. The facial features (eyes) and battery pack were both injection molded plastic. The filling (polypropylene), outside pelt (polyethylene), facial features, and battery pouch (polyester) were all plastic materials. The two AA batteries were produced from multiple components, and we referenced published research (Dolci et al., 2016) to determine the battery composition (Table 3). Both Plush Dogs were packaged in the same way (Table 2) and transported along the same route from Guangzhou, China to Chicago, IL by boat and 18-wheeler in a

Table 1
Each toy's overall mass in grams. Photo credit goes to Hanna Petroski for Lego and Marble Frenzy. Jenga photo by Lizette Arroyo. All other photos by the authors.

Toy	Model Number	Number of Pieces	Weight (grams)	Photo
Jenga	A2120	54	791.94	
Lego Star Wars	75,172	691	373.83	
Marble Frenzy	N/A	50	306.85	
Plush Dog with Battery	N/A	1	203.34	

(*continued on next page*)

Table 1 (*continued*)

Toy	Model Number	Number of Pieces	Weight (grams)	Photo
Barbie	Fashionista 103, Fashionista 111, Fashionista 112		185.00 181.67 180.00	
Plush Dog	N/A	1	152.78	
Lego Cat Woman	70,902	139	147.58	
Lego Batgirl	41,230	206	60.68	

shipping container. The plush dog with batteries was the only toy in this paper with an associated use phase. We assumed the toy needed one change of batteries during its lifetime. We defined the system boundaries of the plush dogs (with and without batteries) in Figs. 1 and 2.

3.6.2. Marble frenzy

Marble Frenzy is a plastic marble track that children can design and rearrange comprised of three forms of plastic. The marbles were made from polyvinylchloride, the bag to hold the marbles was polyethylene terephthalate, and the tubes and miscellaneous game components were polybutylene terephthalate. We assumed that all plastic was produced using injection molding (Table 4). Therefore, we calculated the impacts for the massed materials, as well as injection molding impacts for the mass of each plastic. We assumed transport from Guangzhou, China to Chicago, IL via ocean on a shipping container and an 18-wheeler. The impacts we included are detailed in Fig. 3. The game did not have a use phase impact since no energy was required for play.

3.6.3. Barbie

Due to the 2016 re-design of the Barbie doll to reflect more realistic body types, three dolls representing a subset of these body types were collected and taken apart to identify the material components (Table 5). The three dolls' mass was recorded and the intermediate mass of Barbie #2 (Table 1) was used for all impact cal-

Table 2
Assumptions made to conduct the LCA of the Plush Dogs (with and without a battery), Plush Dogs Material Description.

	Plush Dog without Battery	Plush Dog with Battery
Material and Process	Mass	Mass with Battery
Nylon granulate + injection molding[1]	Eyes: 6.28 g	Battery pack: 87.53 g
Polyester fiber (PET)[2]	Collar, nose, and fabric behind eyes and nose: 2.38 g	Pouch for battery pack: 4.26 g
Polyethylene[1]	Fur coat: 90.27 g	Fur coat: 46.19 g
Polypropylene[2]	Stuffing: 53.85 g	Stuffing: 39.96 g
Transportation		
Guangzhou, China plant to Guangzhou port[3]	53′ truck: 27.35 km	53′ truck: 27.35 km
Guangzhou, China port to Shenzhen, China port[1]	Shipping containers by boat: 153.72 km	Shipping containers by boat: 153.72 km
Shenzhen, China port to Los Angeles, USA port[1]	Shipping containers by boat: 10,306.24 km	Shipping containers by boat: 10,306.24 km
Los Angeles, USA port to Chicago, USA[3]	53′ truck: 3268.58 km	53′ truck: 3268.58 km
Use	No additional energy or resources required once purchased	Battery use over 20 h of entertainment included above

[1] Ecoinvent database.
[2] GabiProfessional database.
[3] National Renewable Energy Lab's Transport database.

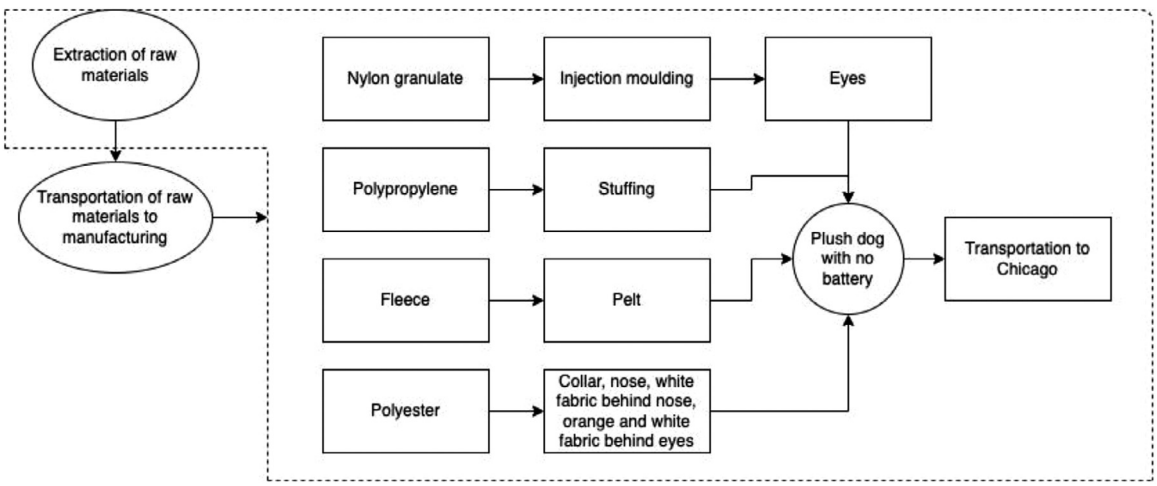


Fig. 1. System boundaries that denote the impacts included (materials, transportation) and excluded (transportation of raw materials to manufacturing plant) for the plush dog with no battery.

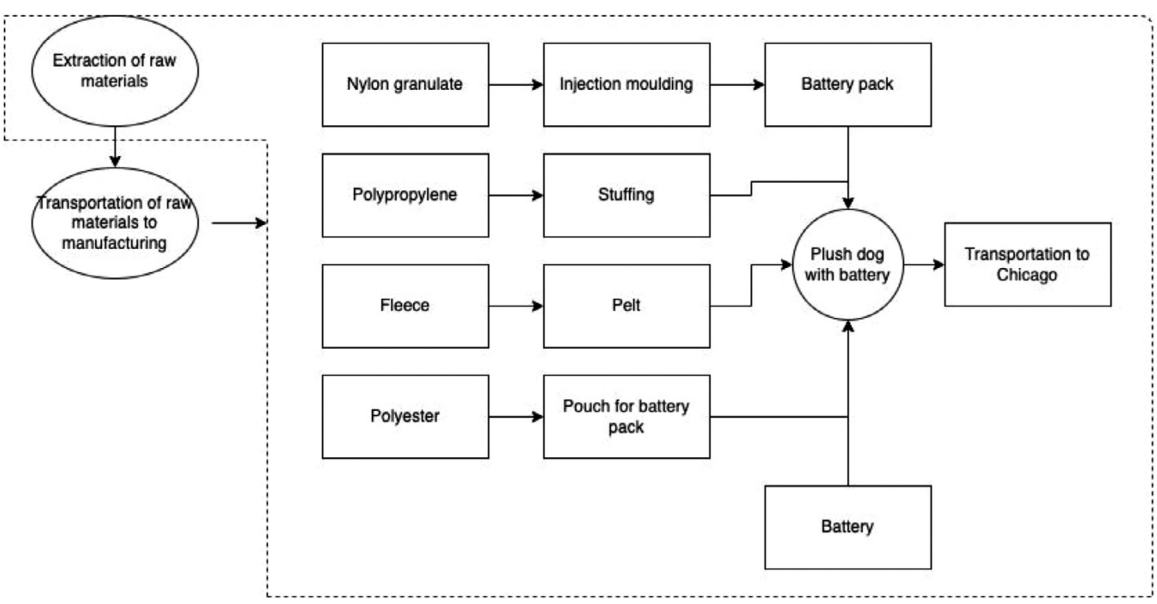


Fig. 2. System boundaries that denote the impacts included (materials, transportation) and excluded (transportation of raw materials to manufacturing plant) for the plush dog with battery.

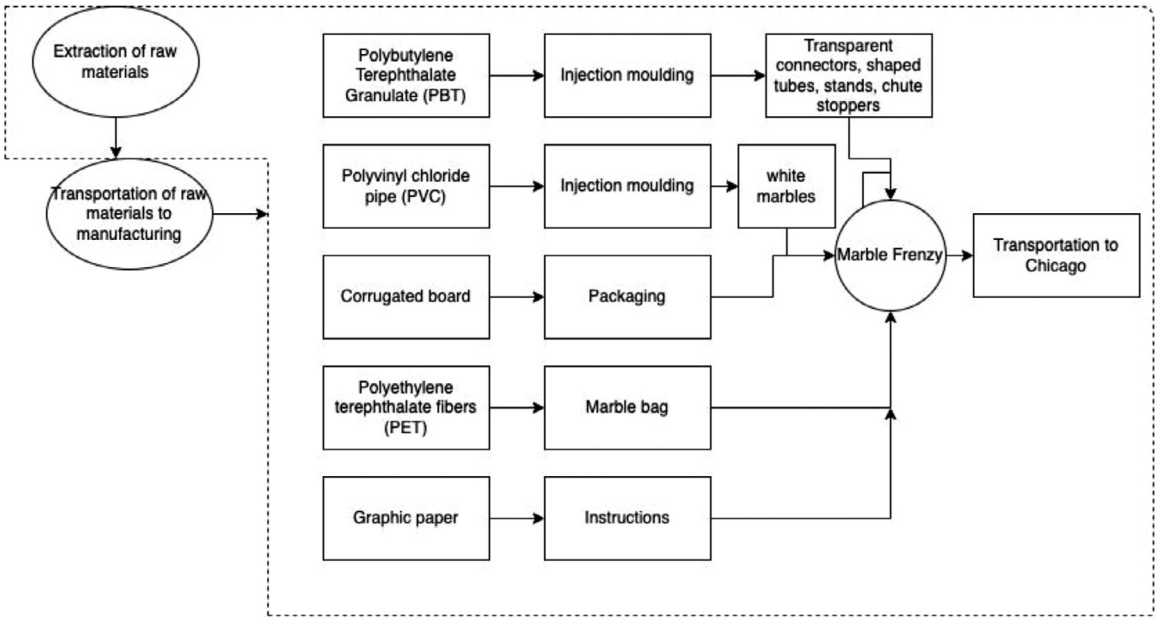


Fig. 3. System boundaries that denote the impacts included (materials, transportation) and excluded (transportation of raw materials to manufacturing plant) for Marble Frenzy.

Table 3
Information used to create the battery for the plush dog with battery (Dolci et al., 2016).

Material and Weight	Percentage of Battery
Manganese dioxide: 8.62 g	37%
Steel: 5.12 g	22%
Zinc oxide: 4.19 g	18%
Potassium hydroxide: 2.56 g	11%
Graphite: 0.93 g	4%
Brass: 0.58 g	2.5%
Carboxymethyl cellulose: 0.35 g	1.5%
Polyvinylchloride: 0.35 g	1.5%
Nylon: 0.35 g	1.5%
Maize starch: 0.23 g	1%

1Ecoinvent database.

Table 4
Assumptions made to conduct the LCA of Marble Frenzy.

Material and Process	Location and Mass
Polybutylene Terephthalate Granulate (PBT)[4] + injection molding[1]	Game pieces (transparent connectors, tubes, stands, and chute stoppers): 161.51 g
Polyethylene terephthalate (PET)[4]	Marble bag: 2.11 g
Polyvinylchloride (PVC)[4] + injection molding[1]	Marbles: 16.5 g
Graphic Paper[2]	Instruction booklet: 1.83 g
Corrugated board[2]	Outer box with dimensions 10.43″ x 10.51″ x 2.52″: 124.9 g
Transportation	
Guangzhou, China plant to Guangzhou port[3]	53′ truck: 27.35 km
Guangzhou, China port to Shenzhen, China port[1]	Shipping containers by boat: 153.72 km
Shenzhen, China port to Los Angeles, USA port[1]	Shipping containers by boat: 10,306.24 km
Los Angeles, USA port to Chicago, USA[3]	53′ truck: 3268.58 km
Use	No additional energy or resources required once purchased

[1] Ecoinvent database.
[2] GabiProfessional database.
[3] National Renewable Energy Lab's Transport database.
[4] GabiPlastics database.

Table 5
Assumptions made to conduct the LCA of a Barbie.

Material and Process	Location and Mass
Polyvinylchloride (PVC)[2]	Head: 55.99 g
Di-isononyl phthalate (DINP)[2]	Head: 13.33 g
Electricity grid mix[2]	Head: 11.66 g
Ethylvinylacetate[2]	Arms: 11.66 g
Acrylonitrile-Butadiene-Styrene Granulate (ABS)[2]	Torso: 22.33 g
Blister packaging[1]	Packaging window: 30.00 g
Linerboard[2]	Outer box: 35.00 g
Transportation	
Guangzhou, China plant to Guangzhou port[3]	53′ truck: 27.35 km
Guangzhou, China port to Shenzhen, China port[1]	Shipping containers by boat: 153.72 km
Shenzhen, China port to Los Angeles, USA port[1]	Shipping containers by boat: 10,306.24 km
Los Angeles, USA port to Chicago, USA[3]	53′ truck: 3268.58 km
Use	No additional energy or resources required once purchased

[1] Ecoinvent database.
[2] GabiProfessional database.
[3] National Renewable Energy Lab's Transport database.

culations. A scenario comparison was done comparing the impact of the high and low masses of the doll. We assumed, based on the plastic type denoted on Barbie's foot that the doll was comprised of three types of plastic (Coda Blog, 2016): the torso was acrylonitrile-butadiene-styrene, the arms made from ethylvinylacetate, and the head and legs from PVC. The head of the Barbie was a soft PVC produced by mixing in an elasticizer (Fig. 4). We used the elasticizer di-isononyl phthalate (DINP) as it is among the most widely used plasticizers, particularly used to make soft and flexible PVC for applications that include toy construction (Park et al., 2008; Wilkinson and Lamb, 1999). We were able to find information on DINP in the GabiProfessional database (Table 5). To add the elasticizer, the PVC's temperature was increased by 35–65 °C (Spielmann and Scholz, 2005; Wilkes et al., 2005). PVC must be heated to 105^{0}C for additives such as plasticizers to be mixed in (Wilkes et al., 2005). The given product system only accounted for

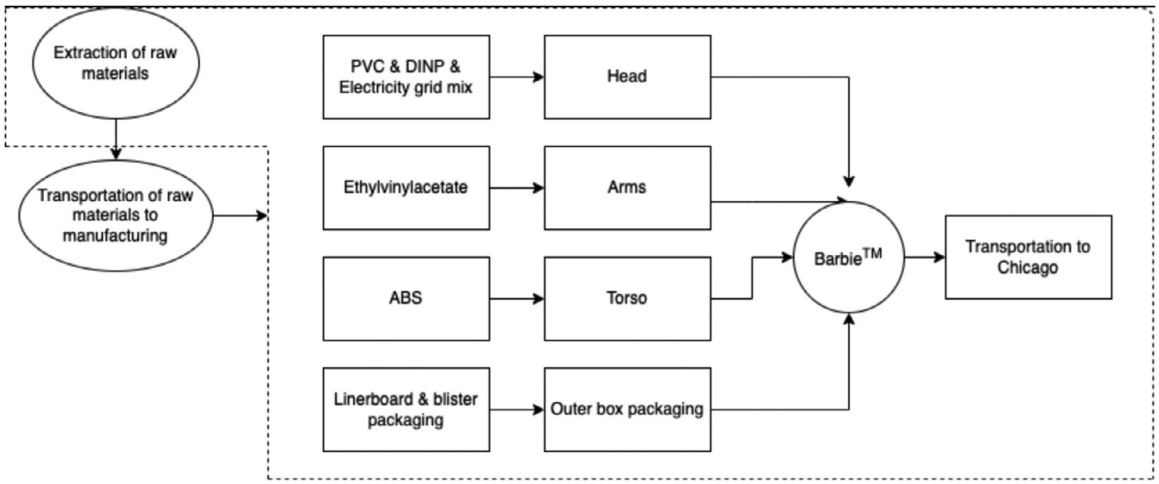


Fig. 4. System boundaries that denote the impacts included (materials, transportation) and excluded (transportation of raw materials to manufacturing plant) for Barbie.

the PVC being heated to 40–70^{0}C (Spielmann and Scholz, 2005). To include the increased temperature, we ran the product system "Electricity grid mix, consumption mix, to consumer, AC, technology mix, <1kV" in the Gabi Professional database to calculate the required electricity to raise the PVC granulates by 35–65^{0}C with a specific heat of 840–1170 J/kg^{0}C (The Engineering ToolBox, 2017). DINP has been the most common plasticizer found in children's toys since the U.S Consumer Product Safety Commission banned di (2-ethylhexyl) phthalate over 0.1% quantity in children's toys (Babich et al., 2020). In 2018, the U.S. Consumer Product Safety Commission banned the use of DINP in children's toys over a presence of 0.1% (Babich et al., 2020). A recent study on chemicals in children's' toys noted that DINP was on the Hazard Index with a warning level of 6.53 × 10° (Aurisano et al., 2021). DINP can make up between 15 and 50% of PVC formulation (Chaudhary et al., 2016). For this calculation we assumed 25% DINP. To determine the impact that different assumptions on DINP content would have on our results, we ran a scenario comparison with 1% and 15% DINP. All of the plastic components for Barbie were assumed to be produced with injection molding and it was assumed the mass of the parts was the mass required for injection molding. We used a combination of five different process flows from GabiProfessional to replicate the production of a Barbie doll (106.66 gs) (Table 5). The first three process flows are specific to the PVC components of the toy (the head and legs) and the other two are for specific components of the toy (torso and arms) made from different plastic types (ABS and EVA, respectively) (Table 5).

We calculated the transportation impacts of the dolls being produced in Guangzhou, China and being transported to Chicago, IL via shipping container and 18-wheeler. Our scenario comparison analysis included changing production location from China to Mexico.

3.6.4. Lego

Three sets of Lego were collected (Catwoman, Batgirl, and Star Wars) and their component materials were identified, massed, and used for impact calculations (Table 6, Fig. 5). Prior to 2019, Lego was made from Acrylonitrile Butadiene Styrene (ABS), however during the research process Lego announced that it had completely transitioned to an eco-friendly bio-plastic for block production (Table 6). Lego made the switch to bio-based blocks due to their overall CO_2 emissions reaching 1 million tons annually (Reed, 2018). Lego was working to reduce this impact by eliminating the plastic bags that hold Lego blocks, and switching to plant-based fibers and recycled bottles by 2030 (Reed, 2018). This bio-plastic is "made out of sustainable sugar cane; The sugarcane [Lego] use[s] is sourced from Brazil with guidance from the Bioplastic Feedstock Alliance (BFA) and is certified by the Bonsucro Chain of Custody Standard for responsibly sourced sugarcane" (Lego, personal communication). We conducted a scenario comparison to determine the impact of using ABS and a bio-based plastic from the Ecoinvent dataset: Bio polyethylene Low Density Granulate (LDPE/PE-LD) (Table 6) (Wernet et al., 2016). This may not be the same bio-based plastic used by Lego. Synthetic rubber was used for all wheel components (Table 6).

During the manufacturing process the plastic was injection molded into bricks. Lego production occurs around the world, however their main production facility is in Billund, Denmark. While Lego recently opened a new factory in China, we did not include production in China in our impact analysis.

3.6.5. Jenga

Jenga was made from Beech hardwood (Table 7, Fig. 6). Stand establishment with planting, site preparation, maintenance of seedlings and forest roads, thinning, and final harvest were included in the beech hardwood production (Berg and Karjalainen, 2003; Berg and Lindholm, 2005; KilpelÄInen et al., 2011). The manufacturing phase of Jenga included the electric consumption required for processing the raw wood material into finished blocks of wood: 67 kgwatt hours cubic meter wood (Bergman, 2011). The game was assumed to come from Mexico to Chicago, IL via an 18-wheeler truck (Table 7).

4. Results

4.1. Greenhouse gas emissions

Our results indicate that the Lego Star Wars set had the highest GHG emissions of all the toys (Fig. 4). Lego Star Wars' impact is due predominantly to its entire mass (373.83 g) being comprised of ABS plastic. In Lego Star Wars, ABS accounted for 64.38% of the toy's overall mass and 78.5% (1.21 kg CO_2 equivalents) of the toy's total GHG emissions (1.54 kg CO_2 equivalents). The plush dog with a battery pack had the second highest GHG emissions (Fig. 7). This was only partly due to the battery impact; the nylon granulate (0.82 kg CO_2 eq) that composed the battery pack (to hold the batteries) was 61.1% of the overall impact (1.33 kg CO_2 eq) (Fig. 7). Jenga had the lowest GHG emissions of all subject toys despite

Table 6
Assumptions made to conduct the LCA of 3 Lego Sets.

Material and Process	Location and Mass Cat Woman Lego Set	Location and Mass Bat Girl Lego Set	Location and Mass Star Wars Lego Set
Acrylonitrile Butadiene Styrene (ABS)[2] + Injection molding[1]	Traditional bricks: 142.11 g	Traditional bricks: 47.21 g	Traditional bricks: 373.83 g
Biopolyethylene Low Density Granulate (LDPE/PE-LD)[2] + Injection molding[1]	Bioplastic bricks: 142.11 g	Bioplastic bricks: 47.21 g	Bioplastic bricks: 373.83 g
Synthetic rubber[1]	Wheels: 4.47 g	Wheels: 13.47 g	No synthetic rubber was used
Liner board[1]	Outer box: ~150 g	Outer box: ~150 g	Outer box: ~220 g
Gloss coated paper[1]	Instruction booklet: 60.36 g	Instruction booklet: 42.73 g	Instruction booklet: 128.05 g
Transportation			
Billund, Denmark plant to Copenhagen, Denmark port[3]	53′ truck: 263.93 km	53′ truck: 263.93 km	53′ truck: 263.93 km
Copenhagen, Denmark port to New York City, USA, port[1]	Shipping containers by boat: 6502.30 km	Shipping containers by boat: 6502.30 km	Shipping containers by boat: 6502.30 km
New York City, USA to Chicago, USA[3]	53′ truck: 1276.85 km	53′ truck: 1276.85 km	53′ truck: 1276.85 km
Use	No additional energy or resources required once purchased	No additional energy or resources required once purchased	No additional energy or resources required once purchased

[1] Ecoinvent database.
[2] GabiProfessional database.
[3] National Renewable Energy Lab's Transport database.

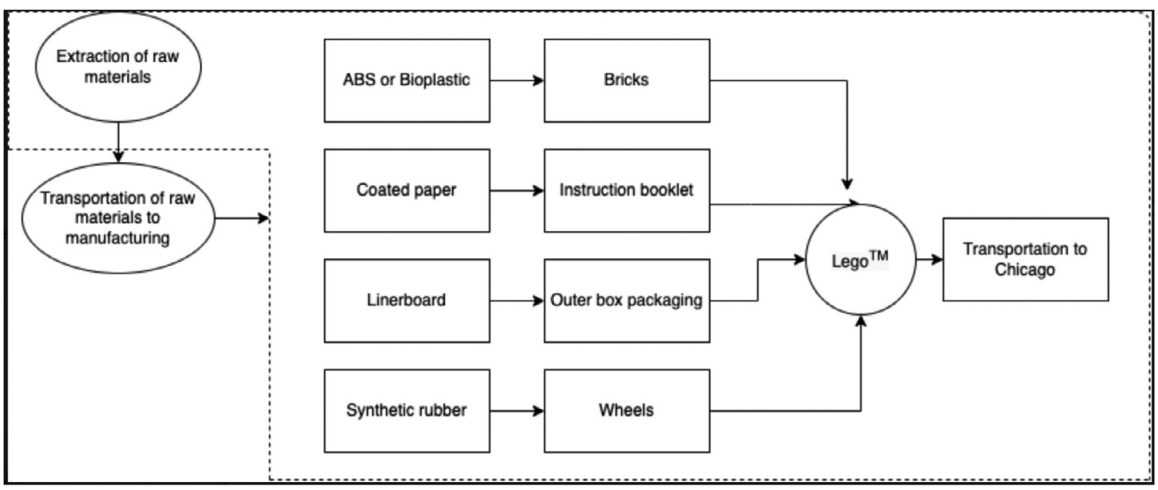


Fig. 5. System boundaries that denote the impacts included (materials, transportation) and excluded (transportation of raw materials to manufacturing plant) for Lego.

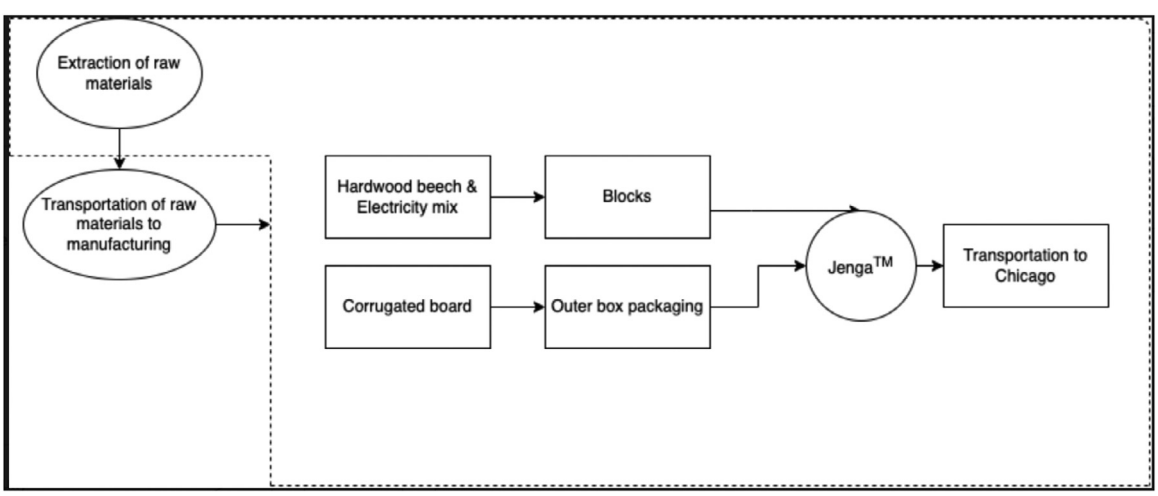


Fig. 6. System boundaries that denote the impacts included (materials, transportation) and excluded (extraction of raw materials, transportation of raw materials to manufacturing plant) for Jenga.

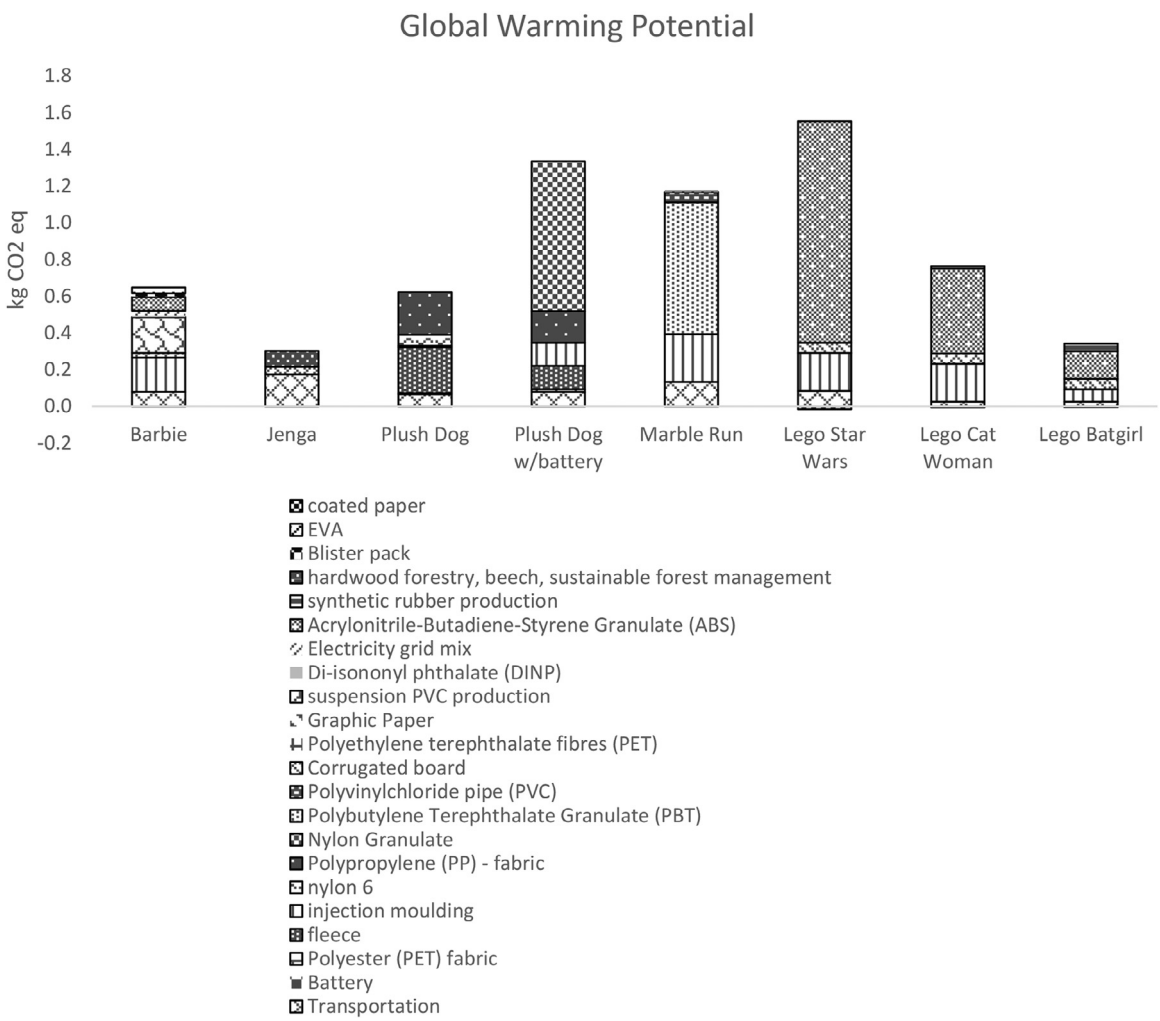


Fig. 7. This graph illustrates each toy's greenhouse gas emissions in kg CO_2 equivalents.

Table 7
Assumptions made to conduct the LCA of a Jenga set.

Material and Process	Location and Mass
Hardwood Beech[1]	Blocks: 732.24 g
Electricity grid mix[2]	Blocks: 0.097 g
Corrugated board[2]	Outer box: 59.7 g
Transportation	
Nuevo Leon, Mexico to Chicago, USA[3]	53′ truck: 2467.12 km
Use	No additional energy or resources required once purchased

[1] Ecoinvent database.
[2] GabiProfessional database.
[3] National Renewable Energy Lab's Transport database.

being the heaviest toy with a mass of 791.94 g (Table 1). Jenga's GHG emissions were only 19.7% of Lego Star Wars' despite weighing roughly twice as much (Fig. 2 and Table 1).

Based on the materials that comprised the toys in this study, plastic had the highest GHG emissions per kilogram. The first and second most impactful materials were nylon granulate (battery pack) (9.31 kg CO_2 eq/1.0 kg nylon granulate) and PET (plush dogs' fur) (6.64 kg CO_2 eq/1.0 kg PET) (Fig. 7), respectively. Polyethylene (Marble Frenzy game pieces) had the third highest GHG emissions with 4.63 kg CO_2 eq per 1.0 kg.

Maize starch, found in the battery, had the lowest GHG emissions of any material with 2.2×10^{-4} kg CO_2 eq/ 1.0 kg maize starch. The 3 materials with the lowest GHG emissions were all present in trace amounts in the battery pack of the plush dog. (SM Table 1). See Table 1 in the supplementary material for a full list of each material in 1.0 kg impact results.

4.2. Eutrophication

Injection molding is the manufacturing process of injecting molten material into a mold. Each of our toys, except for Jenga, had some toy part that was formed using injection molding. Upon inspection of a plastic toy piece, if there are small, indented circles, it is usually a sign that the piece was created through injection molding. This process played the largest role in each toy's eutrophication potential as the injection molding system boundaries included the electricity needed to power the machinery. Creating electricity releases gasses when fuel is burned, including nitrous oxides which leads to eutrophication (Ngatia et al., 2019). Lego Star Wars had the largest Eutrophication potential of all eight toys, with coated paper comprising 53% of the impact. The plush dog with battery, the second most impactful toy in terms of eutrophication potential, had 36.2% of its impact from nylon granulate (Fig. 8). Transportation also played a large role in each toy's eutrophication impact. GHGs, including nitrous oxides, are released into the atmosphere during transportation, contributing to eutrophication. Jenga had the lowest eutrophication potential despite Jenga's transportation impact contributing 94% of the total impact likely due to the toy's high mass.

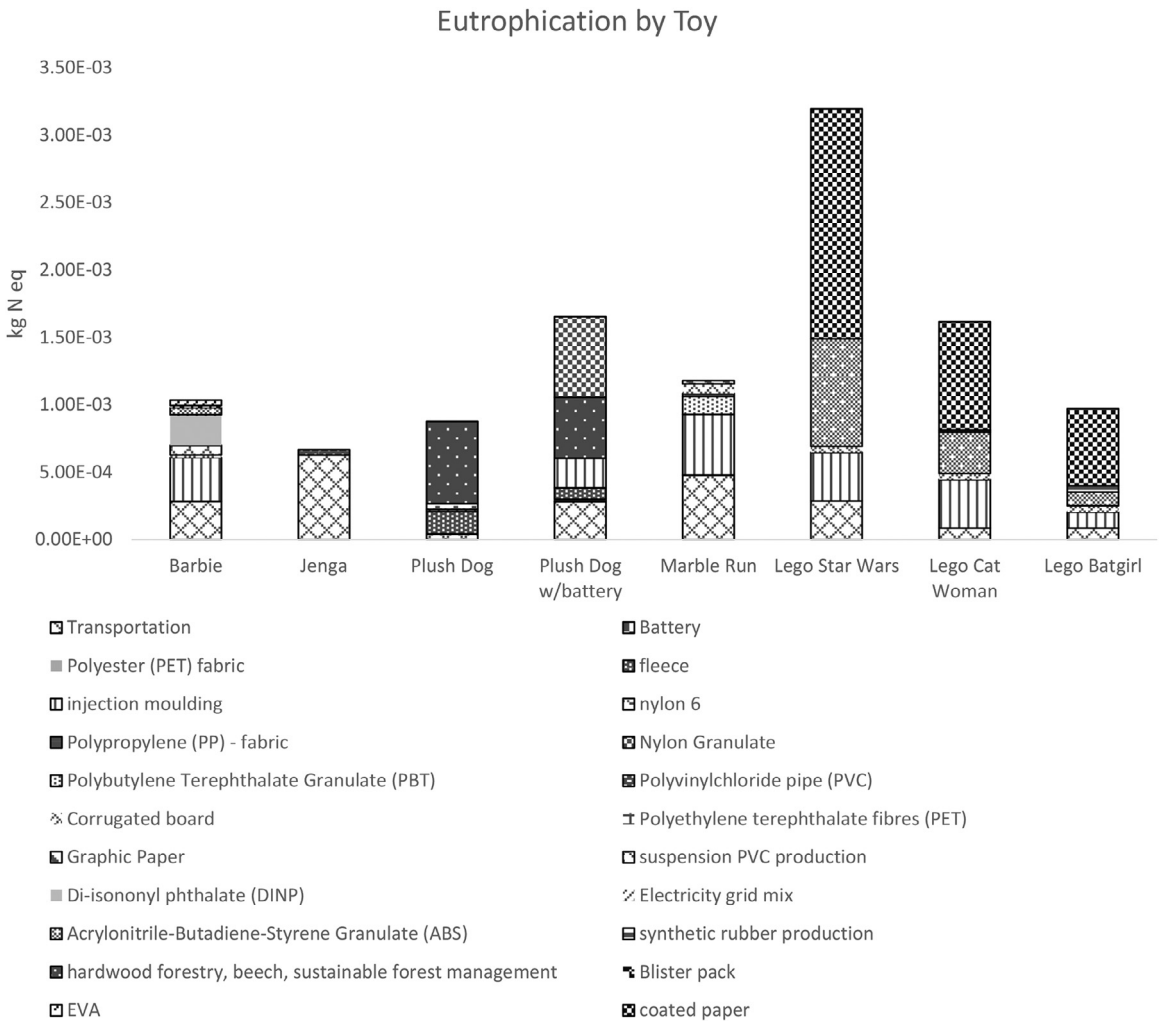


Fig. 8. This graph illustrates each toy's Eutrophication potential in kg N equivalents.

4.3. Acidification

Jenga had the highest acidification potential. This dataset included the deforestation that occurs to source the wood, as well as the heavy machinery needed plant, maintain and havest trees. The plush dog with a battery was the second most impactful in terms of acidification, but not due to the battery (Fig. 9). The battery accounted for 0.2% of the acidification impact while the nylon granulate plastic pieces accounted for 49.9% of the toy's acidification potential.

Plastics had the highest acidification impacts of the toy materials except for hardwood beech. The nylon granulate, used for the plush dog with battery's eyes and battery pack, had the highest impact per 1.0 kg with 0.0297 kg SO_2 eq (SM Table 1). Synthetic rubber, used for Lego Cat Woman and Lego Bat Girl's wheels, had the second highest acidification impacts per 1.0 kg with 0.0149 kg SO_2 eq. (SM Table 1).

4.4. Other impacts

We also looked at impact categories including: ecotoxicity, ozone depletion, photochemical ozone formation and resource depletion- fossil fuels. (Table 8). The Jenga game had the highest impacts in acidification and photochemical ozone formation (SM 2). Lego Star Wars had the highest impacts in eutrophication, greenhouse gas emissions, and resource depletion fossil fuels (SM 1 & 2) due to the larger size of the set compared to the other Lego sets. The plush dog had the largest impact in ozone depletion from the fleece (SM 3) and the largest ecotoxicity impact (SM 1).

4.5. Scenario comparisons

We ran scenario comparisons to take into consideration the variations in mass, material, and production impact for Barbie and Lego.

The most significant impact changes resulted from changing the plastic used in Lego production from ABS to a Bioplastic. This increased the GHG emissions of each Lego set by over 100% and 239% for the Star Wars set, since it had the highest mass (Table 1). Changing the location of Barbie production from China to Mexico, where Mattel is known to produce Barbie (Sethi et al., 2011), reduced the impact across categories from a 7% reduction in GHG emissions to a 25% reduction in acidification. Production in Mexico eliminated boat transit and reduced total vehicle miles travelled by truck.

The multiple "body types" of Barbie resulted in three masses and ratios of material use. We conducted a scenario comparison of the three overall masses with the different material use ratios reflected. As expected, smaller doll masses decreased the doll's impact. Since the "intermediate Barbie" mass was 181.67 gs (Table 1), decreasing the mass to 180 gs had a small, though negative effect (Table 9). Increasing the mass to 185 gs had a slightly more

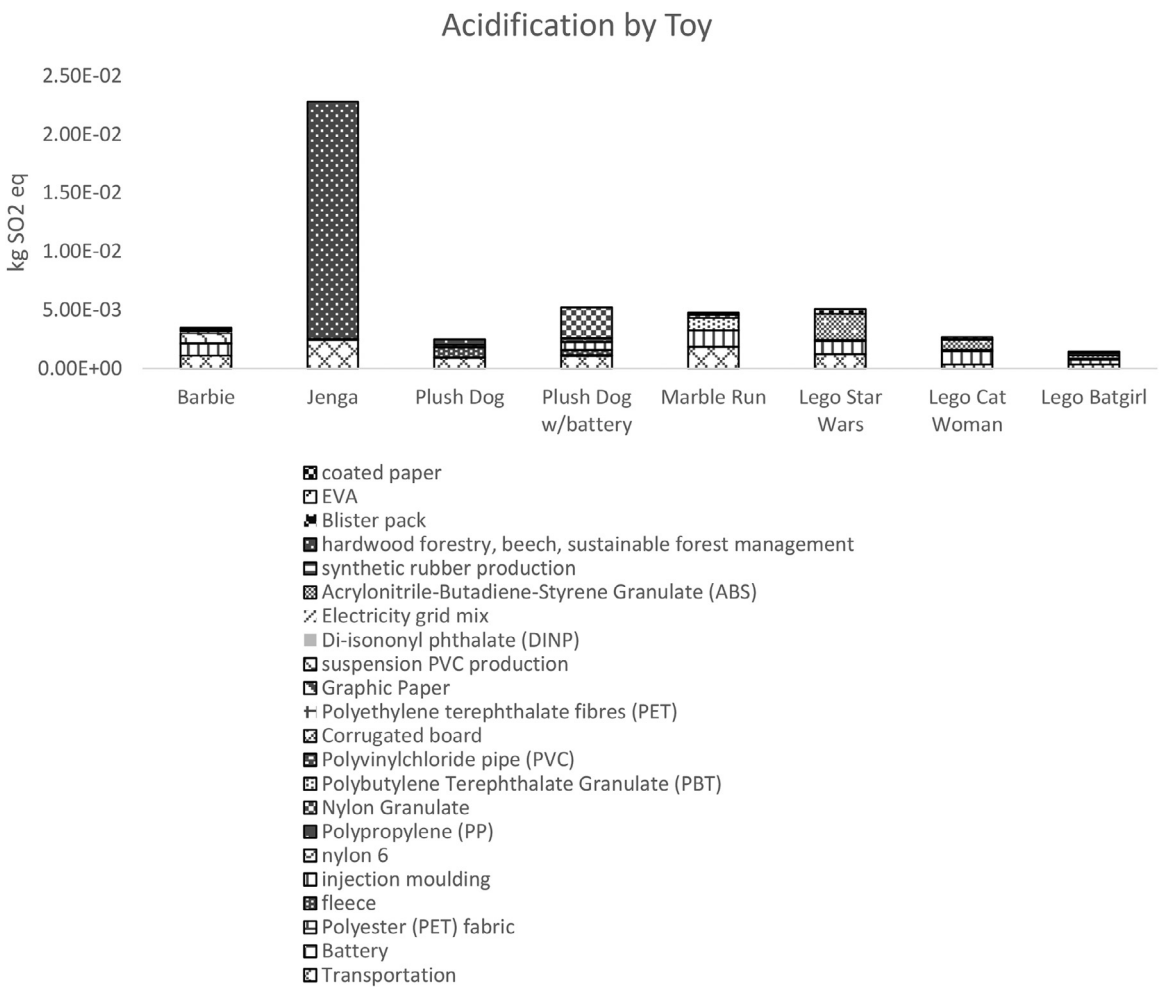


Fig. 9. This graph illustrates each toy's acidification potential in kg SO_2 equivalents.

Table 8
We show the calculated impact of each subject toy in terms of acidification, ecotoxicity, eutrophication, greenhouse gas emissions, ozone depletion, photochemical ozone formation, and resource depletion- fossil fuels. Outlined boxes represent the highest impact in each category.

Toy	Acidification (kg SO2 eq)	Eutrophication (kg N eq)	Photochemical Ozone Formation (kg O3 eq)	Ozone Depletion (kg CFC-11 eq)	Ecotoxicity (CTUe)	Greenhouse Gas emissions (kg CO2 eq)
Plush Dog (Both)	0.0025	0.0009	9.66E-09	9.66E-09	0.0252	0.6226
Plush Dog (Battery)	0.0052	0.0017	1.84E-08	1.84E-08	0.0600	1.3346
Marble Frenzy	0.0048	0.0012	2.97E-08	2.97E-08	0.0517	1.1749
Barbie	0.0035	0.0010	2.19E-08	2.19E-08	0.0307	0.6483
Lego (Star Wars)	0.0051	0.0032	2.05E-08	2.05E-08	0.0174	1.5375
Lego (Batgirl)	0.0015	0.0010	1.68E-08	1.68E-08	0.0057	0.3361
Lego (Cat Woman)	0.0027	0.0016	2.30E-08	2.30E-08	0.0086	0.7554
Jenga	0.0204	0.0000	1.46E-08	2.19E-08	0.0134	0.1278

Table 9
Scenario comparison for Barbie and Lego. The table shows the change in impact given plausible variations in input data.

	Acidification	Eutrophication	Greenhouse Gas emissions
Barbie Transportation from Monterrey, Mexico	−25%	−10%	−7%
DINP 15%	+ <1%	+ <1%	+ <1%
DINP 1%	- <1%	+ <1%	+ <1%
Low Barbie mass	No change	- <1%	+ <1%
High Barbie mass	5%	7%	7%
Star Wars from bioplastic	238%	1506%	239%
Cat Woman from bioplastic	178%	835%	190%
Batgirl from bioplastic	114%	530%	136%

positive effect, in part explained by the slightly larger increase in mass. The material use ratios also contributed to this positive increase. Hard PVC, the highest impact plastic, increased (minimally by 0.008 gs) in this doll while hard PVC decreased (minimally by 0.002 gs) in the lower mass doll. The same was true for EVA with a slight increase of this plastic in the high mass Barbie (increased 0.003 gs) and a decrease of EVA in the lower mass Barbie (decreased 0.002 gs).

While truck transportation is common in the United States, rail is more environmentally friendly. To look at the potential decrease in impacts, we switched all ground transportation in the United States to rail. This switch decreased greenhouse gas emissions, eutrophication, and acidification impacts for most toys. Acidification decreased from 2 to 7%; eutrophication decreased from less than 1% to 107%; and greenhouse gas emissions decreased from 1 to 12% (SM Table 2).

4.6. Material impact per kilogram

Nylon granulate had the highest GHG emissions and acidification impact per 1.0 kg, (SM Table 1). The electricity mix used to soften the DINP in Barbie's head had the lowest GHG emissions, eutrophication, and acidification impact per 1.0 kg.

5. Discussion

In our subset of eight toys, the Star Wars Lego set was the most impactful toy in terms of greenhouse gas emissions and eutrophication potential. Jenga had the lowest GHG emissions, followed by Lego Batgirl, the plush dog, and Barbie. Our results support findings from other research showing the low GHG emissions of wooden toys. A study in India found that locally-made wooden toys were less toxic, consumed less energy, and had "minimum environmental impacts" compared to PVC toys manufactured in China (Rangaswamy et al., 2018). In contrast, preliminary findings on a variety of toys indicated that some plastic toys with a longer life (measured in years of play) had a lower environmental impact (measured in MJ energy use) than the one wooden toy included in the comparison. However, since this was a baby toy only used for 1.5 years, if that toy's lifetime had been doubled (similar to the other lower impact toys), the wooden toy would have been one of the lowest impact toys (Watkins and Mestre, 2019). In addition to finding low GHG emissions for Jenga, this study also provided evidence for the low GHG emissions of some plastic toys. Five of the 8 toys had GHG emissions of less than 1 kg CO_2 equivalents. Four of these were plastic toys (Barbie, the plush dog, and two Lego sets). Indeed, a takeaway for consumer consideration is that lower mass toys (and other materials) can correlate with a lower impact. Both production and transportation impacts depend on the mass of the toy.

The type of plastic used was also important in determining the GHG emissions. Lego's use of ABS and Barbie's predominantly ABS and PVC composition was lower impact than nylon granulate, EVA, and other plastic materials used in toy construction. This indicates the importance of design for sustainability in toy construction (Arnette et al., 2014), though sustainability of materials is often balanced with other factors like plasticity, feel, and look, among others. Indeed, considering the impact hotspots, material choice was critical. For example, nylon granulate, PVC, and PET had higher per kilogram eutrophication than fleece. Interestingly, the use of bioplastic increased the impact of Lego block across impact categories (Table 9), though there are several types of bioplastic from sugarcane (Atiwesh et al., 2021) and whether the system boundaries include land use change can lead to varying results in impact calculation (Bishop et al., 2021). Cardboard and wood were some of the lowest eutrophication potential impacts (SM Table 1), indicating the relatively low impact of toy packaging from these materials. While wood had low GHG emissions and eutrophication potential, it had the highest acidification impact of all materials due to inclusion of forest preparation and stand management in the beech dataset. And when using all impact categories to assess impact, it's clear that there are trade-offs with any material choice (SM Table 1) or toy purchase (Table 8). Indeed, we caution against using the numeric rankings in environmental impact categories as ways to determine best/worst environmentally appropriate toys. The uncertainty around impact rankings is too high to use the results in this way.

There are some results that can inform consumer action. Transportation increased each toy's overall acidification impacts. As fossil fuels are burned and used to power freight, CO_2 and SO_2 are released into the atmosphere and absorbed by the oceans, causing acidification. Oceanic SO_2 makes up 32% of global SO_2 (Gondwe et al., 2003). Transportation impacts were also important in determining a toy's overall impact in the eutrophication and acidification categories, as it contributed between <1% and 39% percent of overall impacts, respectively. While research has found that transportation is not the dominant source of GHG emissions for food production, accounting for only 11% of life-cycle greenhouse gas emissions (Weber and Matthews, 2008), fewer studies have looked at the contribution of transportation to environmental impact in other impact categories. Indeed, transportation was the majority of the eutrophication impact for Jenga. While consumers have no control over the location of manufacturing, they can contribute to reduced impacts from transportation by purchasing lightly-used toys locally. Based on our study results, this choice could provide the most environmental return for high mass toys like Jenga. Our results also indicate the importance of including multiple impact categories, specifically for items like toys, that tend to have a relatively low impact due to their limited material inputs. This may help companies and consumers assess trade-offs between impact categories.

5.1. Circularity and manufactured obsolesce

Since our functional unit depended on the amount of toy use, efforts to extend use could have a significant effect in reducing the environmental impact of each toy. Current toy design practices are not circular, which means most toys end their life cycle in a landfill. A 2014 Chinese study on landfill composition found that 10.62 ± 5.12% of a landfill sample was plastic (Zhou et al., 2014). Of this plastic, 30.87% was non-plastic-bag plastics including: PP, PS, and PVC (Zhou et al., 2014). Plastic toys fall into this landfill category, where they spend over 50 years before they begin to partially, though never fully, degrade (Webb et al., 2012). Overall, the plastics found in each of the toys were more impactful than the wood. However, use of low mass plastics, such as those found in textiles in the plush dogs, show the importance of mass in determining the impact of plastic Impacts increase with the mass of the plastics, during production, transportation, and end-of-life, if not recycled. Conversely, including end-of-life impacts for Jenga would decrease the impact of the toy due to its wooden material. This is significant as the toy industry is dominated by plastics. Ninety percent of the world's toys are made of plastic (Bryant, 2019); The UN Environmental Program found that the toy industry has the highest total direct plastic intensity with 40 tonnes per $1 million revenue (United Nations Environment Program, 2014). More than 70% of the world's toys are made in China and plastic toys account for the majority of Chinese-produced toys (Kang and Zhu, 2015).

Toys, as well as many other consumer goods, are designed under a planned obsolescence framework (Satyro et al., 2018). This started in the toy industry around 1994 when Mattel began using a "rolling mix" where they would release slightly changed prod-

ucts, such as Hot Wheels™ or Barbie, twice over the course of a year (Johnson, 2001). This was done to increase sales: "If customers saw new products every time they went to the store, they were more likely to buy" (Johnson, 2001). When toys are designed with a short life span, their environmental impact will be higher since impact is apportioned over the toy's lifetime. The perception of toys as temporary due to some toys being designed to fail or break due to material choice (Satyro et al., 2018) may also play a role in the constant consumption and purchase of toys, further increasing impact. A 2018 study found that consumers respond negatively to planned obsolescence in design and perceive toys designed for the dump as less valuable (Kuppelwieser et al., 2019), though even with this negative perception, planned obsolescence still increases toy purchases. This is partly due to demand from children. A 2018 study of US based consumers found that 60% of respondents find out about new toys through their children (Statista, 2018), and child toy knowledge often translates into purchase pressure. More transparency and customer education on toy impact might help counteract this purchase pressure from children, though pro-environmental attitudes are not always strongly correlated with pro-environmental behavior (Carrington et al., 2010; De Pelsmacker et al., 2005a; Klimas and Webb, 2018a). Indeed, marketing aimed toward children (Robinson et al., 2001) may counteract any desire for pro-environmental toy purchase behavior on the part of parents.

Pro-environmental behavior is more common when it's the easiest or cheapest option. For example, low-cost choices and actions where the environmental behavior is the default choice increase lower impact choices. (Jaeger et al., 2009; Klimas and Webb, 2018b). Therefore, the role of companies is to help move consumers toward more pro-environmental behavior. Companies are innovating ways to incorporate circular models in the toy industry. Lego Replay™ encourages families to donate used Lego sets to children's charities. U.S.-based Toy-Cycle™ provides infrastructure to send toys back to their manufacturer to be resold. French-based Rejoué™ repairs old toys, and Hong Kong-based Happy Baton™ provides toy subscription services to keep toys in circulation (Robertson-Fall, 2020). Some of the toys included in this study have the benefit of reuse and second lives in Toy Libraries. According to the USA Toy Library Association, there are Toy Libraries in 32 states and counting (U.S. Toy Library Association, 2021). These organizations function as book libraries do, where families can sign up to borrow toys and return them in intervals, to close the loop and prevent toys from reaching the landfill. Participating in programs for toy reuse or recycling is one avenue of action for consumers. Additionally, adults who model conscious consumerism can show children how to value the environment with fewer, lower impact, purchases (Dündar, 2017).

5.2. Consumer choice of toys designed for low impact

The design of toys is also important to consider. Toys that are designed to be reused multiple times (i.e. Jenga), or toys that incorporate parts that can be used in multiple ways, (i.e. Lego), are more sustainable in comparison to toys that are designed only to be used in one specific way with components that cannot be used in other formats (i.e. Marble Frenzy). There are many frameworks to design with sustainability at the forefront: design for assembly (DFA), design for disassembly (DFD), design for serviceability (DFSv), design for quality (DFQ), and design for mass customization (DFMaCu) (Arnette et al., 2014; Clark et al., 2009; Spangenberg et al., 2010). These design methodologies are not limited in scope to toy creation, but when utilized in designing toys, they can reduce the amount of material that ends its lifecycle in a landfill by providing consumers the ability to mend the product and replace parts. Indeed, some of the limited research involving toys has been focused on better design. A case study on the production of a toy train found that material selection, recyclability, and the number of components were important considerations when designing for decreased environmental impact (Choi et al., 1997). Other research on eco-design found impact hot spots in the use phase for an electronic teddy bear. While the paper did not include changes in product design, this research was used to inform manufacturing changes for a similar product (Muñoz et al., 2009). Eco-design and eco-innovation is important from both an environmental and business perspective. A case study in the wine industry found that a sustainability-focused business model can create value to companies, services, and consumers (Serhan and Yannou-Le Bris, 2018). Indeed, sustainable and ethical labels are commonly used to identify environmental and social values in the food industry (Andorfer and Liebe, 2012; De Pelsmacker et al., 2005b; Didier and Lucie, 2008; Garcia-Yi, 2015; Lotade and Loureiro, 2002; Rousseau, 2015), often associated with an increased consumer willingness to pay (Didier and Lucie, 2008; Garcia-Yi, 2015; Klimas and Webb, 2018a).

5.3. Choices to reduce the impact of play

Most toy LCAs have done some form of improvement analysis (Charmondusit et al., 2014; Choi et al., 1997; Margallo et al., 2021) and for good reason. This study's functional unit was 20 h of entertainment over 2 years. All the selected toys provide entertainment in different ways. Plush toys can encourage creativity and imaginative play while blocks like Lego and other building sets were found to inspire more frequent and cooperative social activity (Hendrickson et al., 1981). Games like Jenga are designed for cooperative play. Play with building materials was found to predict reading and mathematics abilities later in school (Wolfgang et al., 2001). Additionally, play quality has been found to vary between toys (Trawick-Smith et al., 2015), so a slight change in functional unit to include educational skill development, or a more specific type of entertainment, could exclude some of the selected toys.

Individual child preferences may lead to variability in the quantity of play for each of the toys. Our functional unit assumes that each toy is utilized for 20 h over two years. Since most of the toys do not have any impact during the use phase, and the initial battery is included with the battery-operated dog, our cradle-use LCA is equivalent to a cradle-gate (consumer purchase) analysis. Duration of play, however, is an important component of toy impact for multiple reasons. Toys that have limited use can be quickly discarded. While this does not change the impact of these toys, it can lead to an increase in toy purchasing. While play is an essential component of child development (Brussoni et al., 2012), the quantity and disposable nature of many children's toys creates a culture of unconscious consumerism from an early age (Chan, 2005). Designing toys for landfill disposal is not necessary to increase enjoyment or play. Toys with extended lifetimes as hand-me downs were preferred in child care settings due their durability (Watkins and Mestre, 2019). Toys are also not a necessary part of play; children often obtain the most health, social, and developmental benefits from free play outdoors (Bento and Dias, 2017; Laaksoharju et al., 2012; Louv, 2008).

While the impact from a single toy item purchase is low relative other household items like appliances (Brommer et al., 2011) that have significant impacts during their use phase, the global impacts of aggregate toy purchases is high. For example, 60 million Barbies were sold annually (Daspin, 2020). Extrapolating from this number, the annual GHG emissions of Barbie production and transportation would be 3390,000,000 kg CO2. If all of these Barbies were made from the same materials as Jenga the GHG emissions would be only 631,000,000E kg CO_2, reducing the impact by 2760,000,000 kg CO_2, the equivalent impact of 310,565,995 gallons

of gasoline (U.S. Environmental Protection Agency, 2020). The impact would be reduced by an even greater amount if fewer consumers purchased the doll. While this decrease in consumption would be countercultural, the number of toys purchased exceeds what is necessary for child development. Indeed, there is a disparity between toy quantity depending on where children live. In Uganda, 75–80% of children between the ages of 3–6 did not have toys (Singla et al., 2015). While absence of toys is an unlikely global scenario for the 92% of children worldwide who live in low and middle income countries (Rebello Britto et al., 2013), the quantity of toys landfilled and dropped at donation centers indicates that consumption exceeds demand in affluent countries and for high-income individuals irrespective of country.

Understanding the impact of toys is important to consider alongside changes in the industry that promote customization. As Barbie and other dolls become more representative of the population, this customization may increase consumption. Indeed, customization increases economic value to the consumer (Franke et al., 2010). Coupling customization with increased toy longevity can help offset these increased impacts, especially since customization increases willingness to pay (Franke et al., 2010) so higher prices could offset environmental impacts while increasing toy longevity.

5.4. Limitations and assumptions

Our study would have benefitted from additional information regarding each material process. Since toy companies typically do not publicly release the exact toy material composition, we made assumptions of toy material composition when we were not able to identify or determine the material. This means that our results do not specifically represent the impact for any of the trademarked toys in this study. While we contacted each of the companies that produce the toys in this study, there were times when the nature of a screening LCA, which relies on available data, forced us to choose a material from a database that was close to the toy material, but not the exact match. For example, the polyvinyl injection molding part chosen for Barbie, was a cradle to gate inventory based in Europe. No Barbie production is known in Europe, so this is likely from a different industry. Additionally, we know from communications with Lego that the sugarcane for their new bioplastic blocks was “sourced from Brazil with guidance from the Bioplastic Feedstock Alliance (BFA) and certified by Bonsucro chain of custody standard for responsibly sourced sugarcane” (Lego, personal communication). There was only one option in GabiPlastics that was a reasonable proxy for this material: biopolyethylene low density granulate (LDPE/PE) production mix from bioethylene base on Brazilian sugar cane (45% slash and burn) granulate. This material, as named, is not listed on the Bonsucro certified green products website.

Our study is also limited by the constraints of a screening LCA that uses existing data that was not specific to region of production. For the construction of Lego blocks, for example, we combined injection molding with ABS plastic granules to create the blocks. However, the injection molding dataset was based on information from different European and Swiss converting companies. The energy mix from this may differ from the energy mix in Billund, Denmark, Nyíregyháza, Hungary and Monterrey, Mexico where block production actually occurs. We also do not know where polymer production occurs for either Lego blocks, Barbie, the Marble Frenzy game, or the plastic used for fleece production in the stuffed dogs. Our assumptions of location of manufacture as well as transportation distances based on those locations also affect our results.

Additionally, as detailed in the section above, individual child preferences may lead to variability in the quantity and time of play with each of the toys selected for this study. Play time may differ by age and the type of entertainment provided by specific toys. Indeed, this was one of the challenges of this study. While toys all provide entertainment, as broadly defined, the type of entertainment could be refined in future studies to focus on toys that develop artistic skills, creativity, or dexterity, among other play categories.

6. Conclusion

This research indicates a low impact per toy for the eight toys analyzed that is comparable with the impact for other forms of entertainment like books, movies (Klimas and Shaffer, 2019) and results from other toys (Rangaswamy et al., 2018; Watkins and Mestre, 2019). In our subset of eight toys, the Star Wars Lego set was the most impactful toy in terms of greenhouse gas emissions and eutrophication potential. Jenga had the lowest GHG emissions, followed by Lego Batgirl, the plush dog, and Barbie. This research illustrates the importance of toy design for the environment, specifically the importance of material choice in determining impact. While we did not include any games or technology dependent toys in this study, we did find that a plush dog with a battery was comparable in GHG emissions to large Lego sets. Creative toys like Lego and Barbie that require imagination can extend toy lifetime and lower impact per hour of use. Consumer choices can also decrease impact. Purchasing second hand wooden toys locally can increase toy longevity and, if not accompanied by additional toy purchases, can reduce the transportation impact associated with high mass toys.

Our results also indicate information that can be used for eco-design and eco-informed consumption. Of the materials that formed the majority of toys, wood had the lowest GHG emissions per mass. These results provide information to consumers that can be used to improve ecologically informed choices. That said, the overproduction and purchase of toys is an issue that cannot be tackled by eco-design alone. Increased toy longevity, even with toys designed for recyclability, cannot address the scale of unnecessary toy purchasing. The information from this study could be combined with surveys on toy purchasing behavior to determine options for reduced consumption of low longevity, high impact toys that are still priced for profit.

Disclaimers

This paper does not represent exact information for all toys as exact material components are copyrighted and not made publicly available. We have made the best estimates for material type based on research but do not claim to have produced an entirely exhaustive or representative model of any of the copyrighted products in this paper.

Supplementary Material

Table 1: We show the impacts per 1 kg material for each of the toy parts in all of our subject toys in terms of: acidification, ecotoxicity, eutrophication, greenhouse gas emissions, ozone depletion, photochemical ozone formation, and resource depletion- fossil fuels from CML Baseline database.

Table 2: We show the impacts of changing all land transportation from truck to rail and the associated change in impact for each of the study toys. All characterization factors were from CML Baseline.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.spc.2022.03.001.

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